

Forecasting Airbnb Booking Activity in Los Angeles with Machine Learning

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This project examines whether machine learning can predict whether a Los Angeles Airbnb listing-night is booked rather than open using a panel constructed from repeated Inside Airbnb snapshots. The analysis builds a structured listing-night dataset from repeated *calendar.csv.gz* and *listings.csv.gz* files and frames the task as a classification problem using a booking proxy, *booked_open_proxy*. Model performance is evaluated under a strict chronological split that uses 2025 for training and 2026 for validation and testing.

The project compares an L1-penalized logistic regression with an XGBoost classifier. The results show that the engineered panel contains meaningful predictive signal and that XGBoost outperforms the logistic baseline on the main ranking and classification metrics. Robustness checks with additional event and weather controls do not overturn this conclusion. However, both models show weak probability calibration, so the predicted probabilities should be interpreted more as relative ranking scores than as precise booking probabilities.

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1. Introduction

This project studies the following research question: can machine learning predict whether a Los Angeles Airbnb listing-night is booked rather than open using a panel built from repeated Inside Airbnb snapshots?¹ The analysis uses repeated *calendar.csv.gz* and *listings.csv.gz* files as its core inputs and applies panel construction and feature engineering to build a structured listing-night dataset. The objective is predictive rather than causal: the aim is to evaluate out-of-sample forecasting performance, not to estimate the effect of any individual factor on bookings.

This question is economically relevant because short-run occupancy conditions affect host revenue, pricing decisions, and broader market conditions in short-term rental markets. Although Inside Airbnb data do not directly record completed bookings, calendar availability still provides useful information about demand. At the same time, because an unavailable night does not necessarily indicate a booking and may instead reflect host blocking, temporary delisting, or other non-booking reasons, the project uses a proxy outcome, *booked_open_proxy*, and restricts the sample to likely active nights, where availability is a more informative signal of booking status.

A key contribution of the project is the construction of a snapshot-based panel that tracks how the availability of the same listing-night changes as the stay date approaches. The analysis compares an L1-penalized logistic regression with an XGBoost classifier under a strict chronological split, using 2025 for training and 2026 for validation and testing, so that the results better reflect seasonality and real forecasting conditions. The results indicate that the engineered panel contains meaningful predictive signal, with XGBoost providing the strongest overall performance, while also highlighting important limitations related to calibration, the proxy-label design, and temporal change over time.

The rest of the report proceeds as follows. Section 2 describes the data, target construction, and feature engineering. Section 3 explains the machine learning framework and the two models used, L1 Logistic and XGBoost. Section 4 presents the results, robustness checks, and limitations. Section 5 concludes.

2. Data and Target Construction

2.1 Airbnb Snapshots

The core data for this project come from repeated Inside Airbnb snapshots for Los Angeles, using both *calendar.csv.gz* and *listings.csv.gz* files. The working scope of the analysis covers listing-night observations from 2025 to 2026, with 2025 used as the historical training window

¹ Repeated Inside Airbnb snapshots are versions of the Airbnb data collected on different points in time. Each snapshot shows listing information and calendar availability as observed on that collection date. They were useful in this project because they made it possible to track how the availability of the same listing-night changed as the stay date got closer, which helped build the panel used to predict booking activity.

and 2026 reserved for forward-looking validation and testing. To keep the modeling pipeline computationally manageable while preserving meaningful cross-sectional variation, the final working sample was restricted to 3,000 listings.

Repeated snapshots are central to the design because they allow the same future stay date to be observed from multiple collection dates. This makes it possible to study how apparent availability evolves as arrival approaches, which is useful for prediction. At the same time, repeated observation creates duplication, since the same (*listing_id*, *date*) pair can appear in more than one snapshot. To construct a usable panel, the snapshot files were merged across collection dates and sorted by *listing_id*, *date*, and *snapshot_date*. For each (*listing_id*, *date*) pair, the latest available snapshot was retained. This produced one final observation per listing-night while preserving information about when that stay date had been observed in the booking cycle through the snapshot timing. The resulting curated panel contains 1,651,366 listing-night observations across 3,000 listings, with no duplicate (*listing_id*, *date*) rows. Its observed date range runs from March 2, 2025 to December 10, 2026. These repeated snapshots were therefore transformed into a single structured panel suitable for chronological prediction.

The resulting panel combines listing-level information with nightly calendar status. Because calendar availability does not directly identify whether a night was truly booked, the project defines a proxy target rather than a true booking outcome. Specifically, the analysis first identifies likely active nights using listing-level activity measures computed from 2025 data: a listing must have at least 30 observed nights, an availability rate above 0.05, and an availability rate below 0.95. This filter is intended to exclude listings that appear mostly inactive or mechanically always available, where booking inference would be less credible. The final binary target is then defined as *booked_open_proxy* = $1 - \text{available}$ for these active nights and left missing otherwise. In the final curated panel, the target is defined for 1,078,516 observations, and the active-night share is 0.6531. This construction makes the prediction task more conservative and better aligned with the project’s objective of forecasting booking-related market activity rather than claiming to observe true realized bookings.

2.2 Feature Engineering

The predictive performance of the project depends not only on the raw Airbnb snapshots, but also on how those raw files were transformed into economically meaningful features. The final baseline feature set combines listing characteristics, geographic information, price measures, calendar timing, seasonality, and panel-based indicators designed to capture how booking conditions evolve over time.

The first group of features describes the listing itself. These include location variables such as *latitude*, *longitude*, and *neighborhood*, along with structural attributes such as *property type*, *room type*, *accommodates*, *bathrooms*, *bedrooms*, *beds*, *minimum-night requirements*, and

amenities count. Additional host and quality-related variables, including *superhost status*, *instant-bookable status*, *number of reviews*, and *review ratings*, were also included because they may capture differences in listing quality, host responsiveness, and guest demand.

A second set of features captures pricing. The project does not rely on calendar-level price because that field is missing in most snapshots. Instead, it extracts a *listing-level base price* from *listings.csv.gz*, which provides more stable coverage across observations. From this, the analysis constructs both a *log price* measure and a *neighborhood-relative price variable*. The neighborhood-relative price is based only on the training-year distribution in order to avoid look-ahead bias and to reflect the idea that booking activity may depend not only on a listing's absolute price, but also on how expensive it is relative to nearby alternatives.

The third group of features captures time patterns in demand. These include *year*, *month*, *week of year*, *day of week*, *weekend indicators*, and *holiday variables*. In addition, the project uses Fourier seasonality terms² to represent smooth within-year seasonal patterns that may not be well captured by calendar dummies alone. These features are intended to reflect recurring variation in travel demand, holiday timing, and short-run occupancy patterns.

One of the most important engineered variables is *days_to_snapshot*, defined as the number of days between the snapshot date and the stay date. This variable measures how far in advance a listing-night is being observed. It is especially important in a repeated-snapshot setting because a night observed 200 days before arrival is not directly comparable to a night observed only a few days ahead. Availability observed far in advance may simply reflect that there was still time for the night to be booked, whereas availability observed close to the stay date may be a stronger signal of lower realized demand. Including *days_to_snapshot* helped the model account for this difference. The feature set also includes *listing_hist_booking_rate*,³ which summarizes how often a listing appeared booked in the historical training data and provides a useful signal of cross-sectional demand differences across listings. To avoid target leakage, the model excludes variables that directly reveal or help define the outcome, including *available*, *booked_open_proxy*, *is_actively_open_night*, *listing_available_rate*, and *listing_obs_count*.

Finally, the baseline model includes holiday controls, while weather and event variables are reserved for separate robustness checks. The weather controls were merged as date-level measures of temperature and precipitation. The event controls were engineered as daily indicators of local event intensity, including sports, music, city-calendar, and permit-related activity, using raw Los Angeles city-calendar data, temporary special event permit records,

² First- and second-harmonic sine and cosine terms were used to capture both broad and more flexible seasonal patterns over the year. Unlike simple month indicators, these terms treated the calendar as cyclical, so nearby dates across calendar-year boundaries were represented as close rather than unrelated.

³ This variable was useful because some listings consistently attract more demand than others. A listing's historical booking rate helps capture persistent differences in popularity that may not be fully reflected in the other observed features.

Ticketmaster event data, and sports or match schedule files.⁴ More on the event-control robustness checks is explained in subsequent sections. Overall, the feature-engineering strategy balanced economic interpretability, predictive usefulness, and safeguards against look-ahead bias and target leakage.

3. Application of Machine Learning Tools

3.1 Modeling Framework

The outcome, *booked_open_proxy*, is binary: for each active listing-night, the target indicates whether the night is classified as booked or open. Since the objective is to predict a booked-versus-open proxy rather than a continuous outcome, classification methods are the appropriate modeling tools. To reflect the forecasting nature of the problem, the data were split chronologically rather than randomly. All target-defined observations from 2025 were used for training, while 2026 was reserved for out-of-sample evaluation. The 2026 observations were then divided by date into validation and test sets, with the first half used for validation and the second half used for testing. This design avoids look-ahead bias and provides a more credible assessment of predictive performance under changing market conditions.

Model selection within the training period used expanding-window backtesting over 2025 dates. This approach preserves temporal order by repeatedly training on earlier observations and validating on later ones, making it better suited than random cross-validation for a forward-looking prediction task. Both models were estimated using a common preprocessing pipeline: missing numeric values were imputed with medians, categorical variables were imputed with the most frequent category and one-hot encoded, numeric features were clipped using training-data quantiles, and variables that were entirely missing in training were dropped. For logistic regression, numeric predictors were also standardized.

The analysis compares two methods: an L1-penalized logistic regression and an XGBoost classifier. The logistic model serves as a sparse, interpretable baseline, while XGBoost provides a more flexible alternative that can capture nonlinearities and interactions. Hyperparameters were selected using expanding-window validation performance, and final classification thresholds were chosen on the validation set by maximizing the F1 score. Overall, this framework was designed to compare the two models under a consistent, chronologically valid, and practically relevant forecasting setup.

⁴ These event controls were constructed by cleaning and standardizing the raw event records, converting them to a common daily date format, aggregating them by date, and creating daily event-count and event-day variables for categories such as *sports*, *music*, *city-calendar*, and *permit-related activity*. The resulting date-level features were then aligned to the project’s modeling window and merged into the panel by date.

3.2 Models Implemented

Model 1: L1-penalized Logistic Regression

For listing-night i , the model estimates:

$$P(Y_i = 1 \mid X_i) = \frac{1}{1 + \exp(-X_i' \beta)}$$

Where $P(Y_i = 1 \mid X_i)$ is the predicted probability that listing-night i is classified as booked rather than open: $Y_i = 1$ if *booked_open_proxy* = 1, and X_i denote the vector of observed features for that listing-night.

In this project, the coefficient vector β is estimated using an L1 penalty. The fitted model solves:

$$\hat{\beta} = \arg \min_{\beta} \left[- \sum_{i=1}^N (Y_i \log p_i + (1 - Y_i) \log(1 - p_i)) + \lambda \sum_{j=1}^K |\beta_j| \right]$$

where $p_i = P(Y_i = 1 \mid X_i)$. The L1 penalty shrinks less informative coefficients toward zero, which helps control complexity and improves interpretability.

Model 2: XGBoost Classifier

The second model is an XGBoost classifier, a tree-based boosting method that combines many decision trees to improve prediction. For listing-night i , the model forms a score:

$$\hat{s}_i = \sum_{m=1}^M f_m(X_i)$$

where each $f_m(X_i)$ is a decision tree, and the combined score is converted into a predicted probability using the logistic transformation:

$$P(Y_i = 1 \mid X_i) = \frac{1}{1 + \exp(-\hat{s}_i)}$$

XGBoost is used to capture nonlinear relationships and interactions among features such as price, seasonality, listing characteristics, and *days_to_snapshot*, which may be difficult to represent in a linear model.

3.3 Model Evaluation and Comparison Strategy

Model performance was evaluated using both ranking and classification metrics. The main ranking metrics were ROC AUC and PR AUC. ROC AUC measures how well a model distinguishes booked from open listing-nights across thresholds, while PR AUC focuses on the tradeoff between precision and recall for the positive class. In this project, PR AUC is especially informative because the goal is to identify listing-nights that are more likely to be booked.

Classification performance was also assessed using precision, recall, and F1 score. Precision measures the share of listing-nights predicted as booked that were actually classified as booked under the proxy target, while recall measures the share of booked nights correctly identified by the model. The F1 score summarizes the balance between precision and recall and was used to select the final classification threshold on the validation set, since the default cutoff of 0.5 is not necessarily appropriate when class balance changes over time. The evaluation also includes the Brier score, which measures the accuracy of predicted probabilities. Lower Brier scores indicate better calibration. This matters because a model may rank listing-nights well while still producing poorly calibrated probabilities.

The comparison between models is based on the same chronological data split, preprocessing pipeline, and evaluation metrics. This makes performance differences easier to interpret as differences in modeling approach rather than differences in data treatment. In assessing overall model performance, the greatest weight is placed on PR AUC, ROC AUC, and F1 score, while the Brier score is used as a diagnostic check on probability calibration. All headline results are reported on the held-out test set after tuning and threshold selection were completed using only the training and validation periods.

4. Analysis and Insights

4.1 Interpretation of Results and Model Comparison

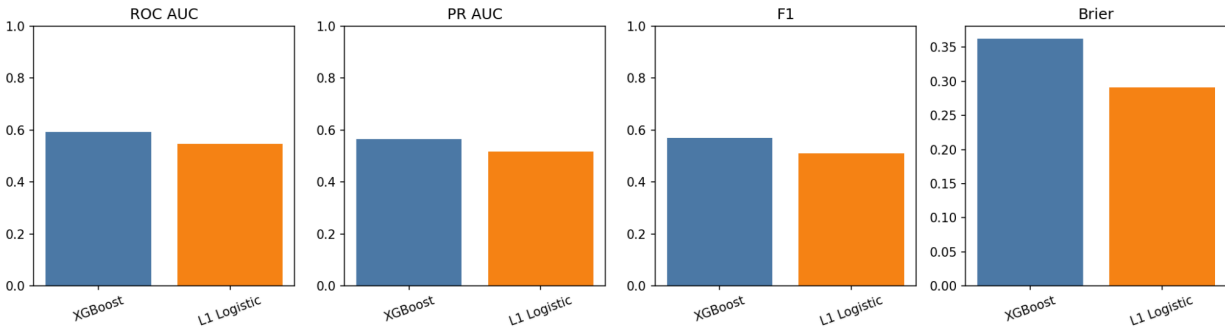
Table 1 and Figure 1 summarize the baseline test-set results. The baseline results show that both models contain some predictive signal, but XGBoost performs more strongly than the L1-logistic benchmark on the main ranking and classification metrics. On the validation set, XGBoost achieved a ROC AUC of 0.7376, a PR AUC of 0.6022, and an F1 score of 0.5969, compared with 0.6465, 0.5046, and 0.5169 for logistic regression. This gap indicates that the more flexible model was better able to distinguish listing-nights that were more likely to be classified as

booked and to convert those rankings into stronger classification performance at the selected threshold.

Table 1. Headline Test Metrics

Model	ROC AUC	PR AUC	F1	Precision	Recall	Brier	Threshold
XGBoost	0.594	0.567	0.570	0.518	0.632	0.363	0.34
L1 Logistic	0.548	0.518	0.510	0.495	0.525	0.291	0.36

Figure 1. Model Performance Panel (Test)



The same pattern remains on the held-out test set, although performance declines for both models. As shown in Table 1, logistic regression recorded a test ROC AUC of 0.5480, a PR AUC of 0.5183, and an F1 score of 0.5095. XGBoost improved on each of these measures, with a test ROC AUC of 0.5936, a PR AUC of 0.5666, and an F1 score of 0.5696. Figure 1 reinforces this comparison by showing that XGBoost outperforms the logistic benchmark across the main test metrics. In practical terms, this means that XGBoost was better both at ranking listing-nights by booked likelihood and at classifying booked-versus-open nights using the validation-selected threshold. For this reason, XGBoost is the preferred final model.

Figure 2 and Figure 3 provide the same conclusion in graphical form. In both figures the XGBoost curve dominates the logistic benchmark on the test set, indicating stronger discrimination across thresholds. The precision-recall comparison is especially important here because the project is primarily concerned with identifying listing-nights that are more likely to be booked rather than simply maximizing overall accuracy.

Figure 2. Precision-Recall Curves (Test)

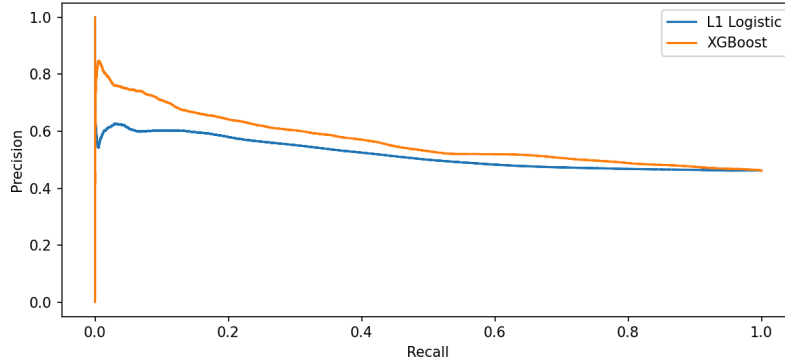
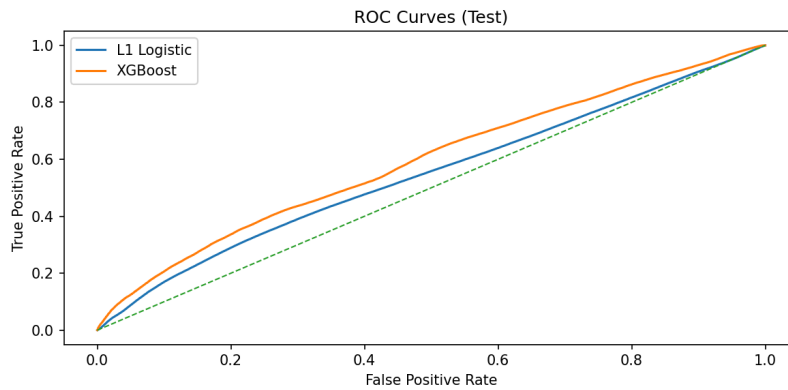


Figure 3. ROC Curves (Test)



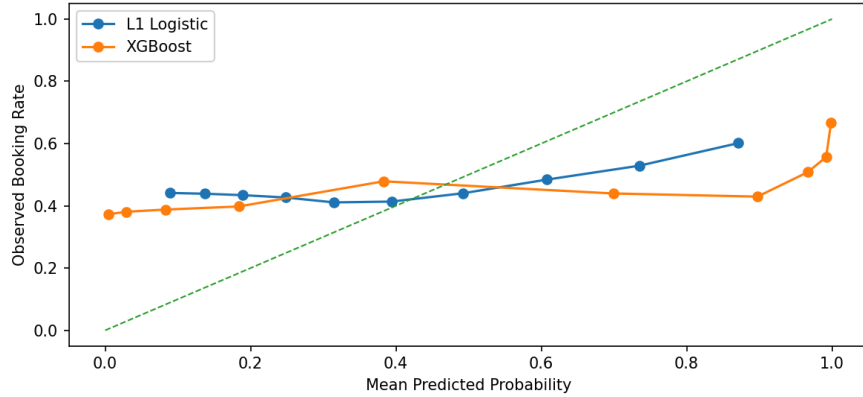
At the same time, the decline from validation to test performance is substantively important. Because the evaluation design is strictly chronological, lower test performance should not be interpreted simply as model failure. Instead, it is more consistent with temporal change in the prediction environment, including shifts in class balance between the training, validation, and test periods. This reinforces the value of the chronological evaluation strategy: a random split would likely have produced more optimistic but less realistic estimates of out-of-sample performance.

The logistic model remains useful despite weaker performance. As an L1-penalized linear benchmark, it provides a sparse and interpretable point of comparison and helps show that the performance gains from XGBoost come from allowing more flexible nonlinearities and interactions rather than from differences in sample construction or evaluation design. In that sense, the comparison is informative not only because XGBoost performs better, but also because it shows that the repeated-snapshot panel contains predictive structure that is not fully captured by a simple linear specification.

Finally, the probability-quality metrics call for caution. On the validation set, XGBoost also had the better Brier score, but on the test set its Brier score deteriorated to 0.3625, compared with

0.2907 for logistic regression. This means that although XGBoost is the strongest model for ranking and threshold-based classification, its predicted probabilities are not well calibrated out of sample. Figure 4 can be used to illustrate this point by comparing predicted and observed booking rates across probability deciles. The results should therefore be interpreted as evidence that the model can identify relatively higher- and lower-booking-risk listing-nights, not as evidence that it produces precise booking probabilities. This distinction is especially important given that the outcome is a proxy target rather than a directly observed booking measure.

Figure 4. Calibration by Prediction Decile (Test)



Additional supporting figures and tables, including confusion matrices and related diagnostic outputs, are provided in Appendix A.

4.2 Robustness Checks

The robustness analysis examines whether the main ranking conclusion changes when the baseline feature set is augmented with additional event or weather controls. The event-controls exercise adds date-level measures of local event intensity, including *event_count_total*, *event_count_sports*, *event_count_music*, *event_count_city_calendar*, and *event_count_tse_permit*. These variables come from a processed daily event-proxy file constructed within the project workflow from Los Angeles city-calendar data, temporary special event permit records, and Ticketmaster event pulls.⁵ The weather-controls exercise adds *temp_max*, *temp_min*, and *precipitation_sum*. In both cases, the purpose is not to redefine the baseline model, but to test whether the main result depends on omitting potentially relevant short-run demand shifters that could affect booking activity.

Table 2 summarizes how test-set performance changes relative to the baseline specification. For XGBoost, the main result is highly stable. Adding event controls changes test ROC AUC from

⁵ These event proxies were built from raw Los Angeles event-data files, including LA city event listings (*la_city_events_2022_2025.csv* and *la_city_events_refresh_2026-03-08.csv*), Los Angeles temporary special event permit records (*la_temporary_special_event_permits.csv*), and Ticketmaster event data (*ticketmaster_los_angeles_*.json*).

0.5936 to 0.5902, PR AUC from 0.5666 to 0.5663, and F1 from 0.5696 to 0.5592. Adding weather controls produces similarly small changes, with ROC AUC at 0.5931, PR AUC at 0.5642, and F1 at 0.5673. These differences are modest and do not overturn the conclusion that XGBoost is the strongest overall model for ranking and classifying the booked-versus-open proxy. More broadly, this suggests that the main predictive signal comes primarily from the baseline panel structure and engineered listing- and timing-related features rather than from these additional controls alone.

Table 2. Robustness Metric Deltas

Scenario	Model	Metric	Baseline	With Controls	Change
Event Controls	L1 Logistic	ROC AUC	0.54795	0.54806	0.00011
Event Controls	L1 Logistic	PR AUC	0.51829	0.51843	0.00014
Event Controls	L1 Logistic	F1 Score	0.50953	0.59474	0.08521
Event Controls	L1 Logistic	Brier Score	0.29074	0.29244	0.00170
Event Controls	XGBoost	ROC AUC	0.59360	0.59024	-0.00336
Event Controls	XGBoost	PR AUC	0.56663	0.56628	-0.00035
Event Controls	XGBoost	F1 Score	0.56960	0.55923	-0.01038
Event Controls	XGBoost	Brier Score	0.36255	0.35641	-0.00614
Weather Controls	L1 Logistic	ROC AUC	0.54795	0.54771	-0.00024
Weather Controls	L1 Logistic	PR AUC	0.51829	0.51808	-0.00020
Weather Controls	L1 Logistic	F1 Score	0.50953	0.50915	-0.00039
Weather Controls	L1 Logistic	Brier Score	0.29074	0.29094	0.00021
Weather Controls	XGBoost	ROC AUC	0.59360	0.59311	-0.00048
Weather Controls	XGBoost	PR AUC	0.56663	0.56416	-0.00247
Weather Controls	XGBoost	F1 Score	0.56960	0.56728	-0.00232
Weather Controls	XGBoost	Brier Score	0.36255	0.35874	-0.00380

The logistic model shows a different pattern under the event-controls robustness check. Its ranking metrics remain essentially unchanged, with test ROC AUC moving from 0.5480 to 0.5481 and PR AUC from 0.5183 to 0.5184. However, its F1 score rises from 0.5095 to 0.5947. According to the notebook outputs, this increase is accompanied by a large threshold shift from 0.36 in the baseline model to 0.14 in the event-controls specification, along with much higher recall. This suggests that the improvement reflects a change in the operating threshold rather than a meaningful increase in the model's underlying ranking ability. For that reason, the event-controls result for logistic regression should be interpreted cautiously.

The weather-controls robustness check is more straightforward. For both models, the weather-augmented specification produces only negligible changes in the main test metrics. This suggests that the baseline feature set already captures most of the predictive structure available at this level of weather aggregation. At the same time, these results should not be read as evidence that event or weather conditions are unimportant in principle. Rather, they suggest that the relatively coarse controls available in the current notebook do not materially alter the main ranking result. If richer data were available, such as more localized event exposure, neighborhood-specific attendance measures, or higher-frequency weather conditions, the models

might capture additional short-run demand variation. This implies that the methodology may be especially useful in settings where booking activity responds strongly to local, time-varying shocks and those shocks can be measured with greater precision.

Taken together, the two robustness exercises support the same overall conclusion: the baseline XGBoost result is reasonably stable, and the project’s main finding does not depend on one fragile specification choice. Additional supporting diagnostics are available in Appendix A. These include the event-robustness and weather-robustness diagnostic tables, which summarize signal availability across splits and provide baseline-versus-robustness metric comparisons in more detail.

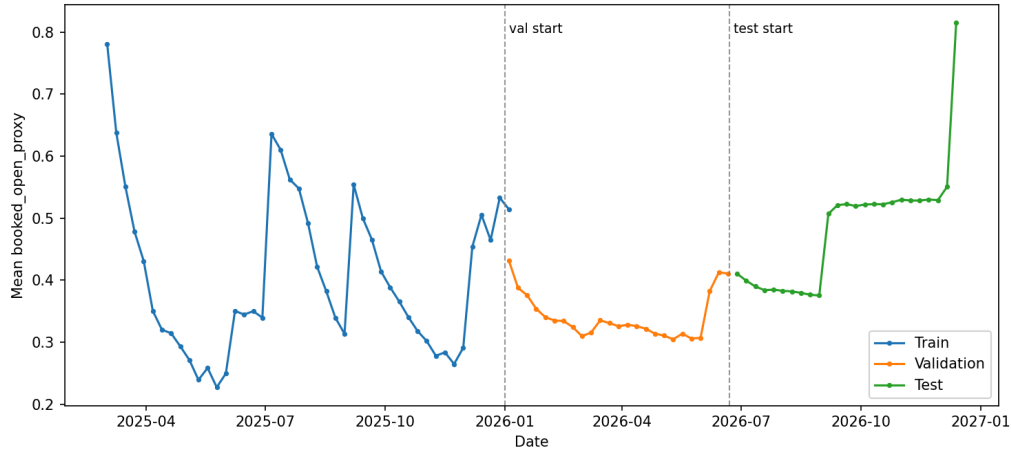
4.3 Limitations

Despite the encouraging predictive results, the project has several important limitations. The first is the outcome itself. The target, *booked_open_proxy*, is a proxy rather than a directly observed booking measure, since calendar unavailability does not necessarily correspond to a realized reservation. As noted earlier, some unavailable nights may instead reflect host blocking, temporary delisting, maintenance, or other non-booking reasons. The active-night filter helps make the booked-versus-open interpretation more credible, but it does not eliminate this source of measurement error. For that reason, the results should be interpreted as evidence about predicting a booking-related proxy, not true realized demand.

A second limitation concerns probability calibration. Although XGBoost performs best on the main ranking and classification metrics, both models show weaker calibration than their discrimination performance alone would suggest. This is especially clear in the test-set Brier scores and in the calibration diagnostics, which indicate that predicted probabilities do not map cleanly into observed booking rates out of sample. As a result, the models are more useful for ranking listing-nights by relative booking likelihood than for assigning precise booking probabilities.

The decline from validation to test performance also points to temporal instability in the prediction environment. Because the evaluation design is strictly chronological, this drop is informative rather than incidental: it suggests that relationships learned from earlier periods do not transfer perfectly to later ones. Figure 5 reinforces this point by showing that the share of listing-nights classified as booked varies across the train, validation, and test periods. Changes in class balance, market conditions, or platform behavior may all contribute to this pattern. In that sense, the project’s forecasting design is a strength, but it also reveals an important limitation of model stability over time.

Figure 5. Target Rate Over Time by Split



The scope of the available data is also an important limitation. Although the project uses a rich set of listing, timing, seasonal, holiday, weather, and event-related features, several potentially important determinants of booking activity are either unavailable or only imperfectly measured. For example, the analysis does not observe realized transaction prices, dynamic pricing adjustments at the nightly level, competitor prices for nearby listings on the same date, search visibility on the platform, host response behavior, or richer neighborhood-level demand conditions. More detailed and localized event exposure, higher-frequency weather measures, and broader market-level demand indicators could also improve predictive performance. As a result, the current models likely capture only part of the variation in short-run booking activity.

Finally, this is a predictive project rather than a causal research design, which is important for interpreting the project’s research question. The project asks whether machine learning can predict whether a Los Angeles Airbnb listing-night is booked rather than open using a panel built from repeated Inside Airbnb snapshots, not whether any one observed feature causally determines booking activity. Variables such as price, listing quality, seasonality, and event intensity may improve prediction because they are associated with underlying demand conditions, even if changing those variables would not by itself produce the same change in bookings. The results therefore answer the forecasting question at the center of the project, but they should not be interpreted as evidence about the causal effect of individual features on Airbnb demand.

5. Conclusion

The project shows that a panel built from repeated Inside Airbnb snapshots contains meaningful predictive signal for short-run booking-related activity in Los Angeles. Under a strict chronological evaluation design, XGBoost outperforms the L1-logistic benchmark on the main ranking and classification metrics, making it the preferred final model. At the same time, the results should be interpreted carefully: the outcome is a booked-versus-open proxy rather than a

directly observed reservation measure, and the predicted probabilities are not well calibrated out of sample.

These findings suggest that short-run Airbnb demand is not purely idiosyncratic, but instead reflects patterns that can be forecast using listing characteristics, pricing information, timing variables, seasonality, and snapshot-horizon measures. For hosts, property managers, or platform analysts, a framework of this kind may be useful for ranking listing-nights by relative booking likelihood and identifying periods of weaker demand. However, because this is a predictive rather than causal exercise, the results should not be interpreted as evidence that changing a single observed feature would necessarily cause bookings to change in the same way, nor should they be used on their own to draw policy conclusions about regulation or market interventions.

Several next steps would realistically strengthen the analysis. Richer demand-side variables, such as competitor pricing, platform visibility, neighborhood-level market conditions, and more localized event exposure, could improve performance. Better calibration would make the predictions more useful as probabilities rather than relative ranking scores. Expanding the analysis across a longer time horizon and additional cities would also help assess how stable these predictive relationships are over time and across markets. Overall, this project contributes a reproducible, chronologically evaluated forecasting framework while remaining appropriately cautious about measurement, calibration, and causal interpretation.

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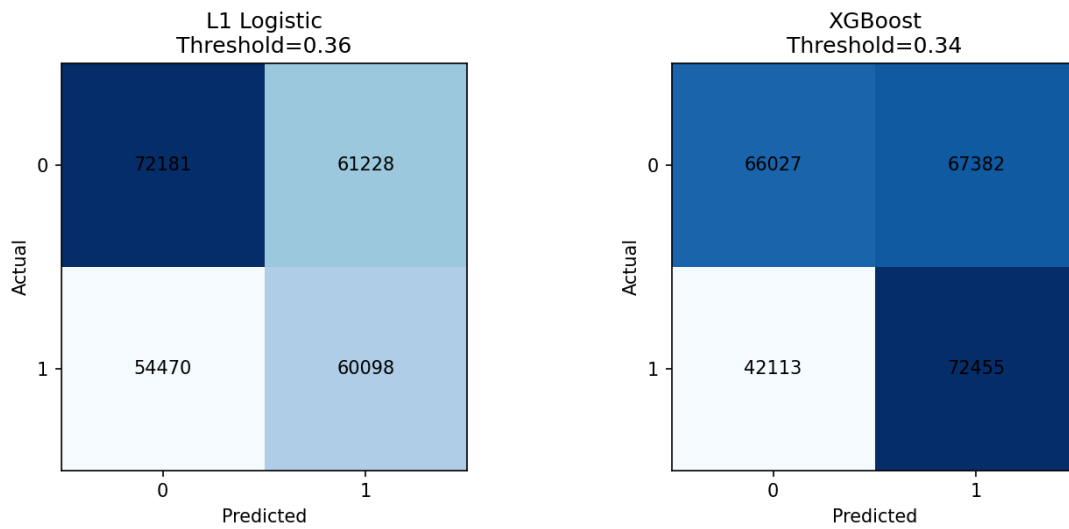
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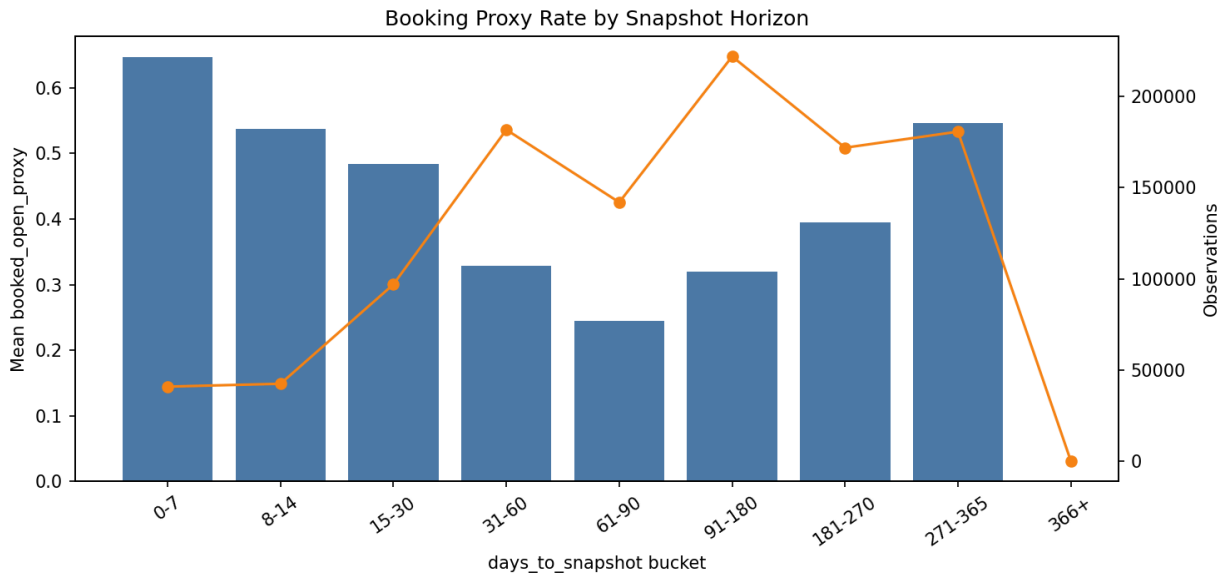
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Appendix A

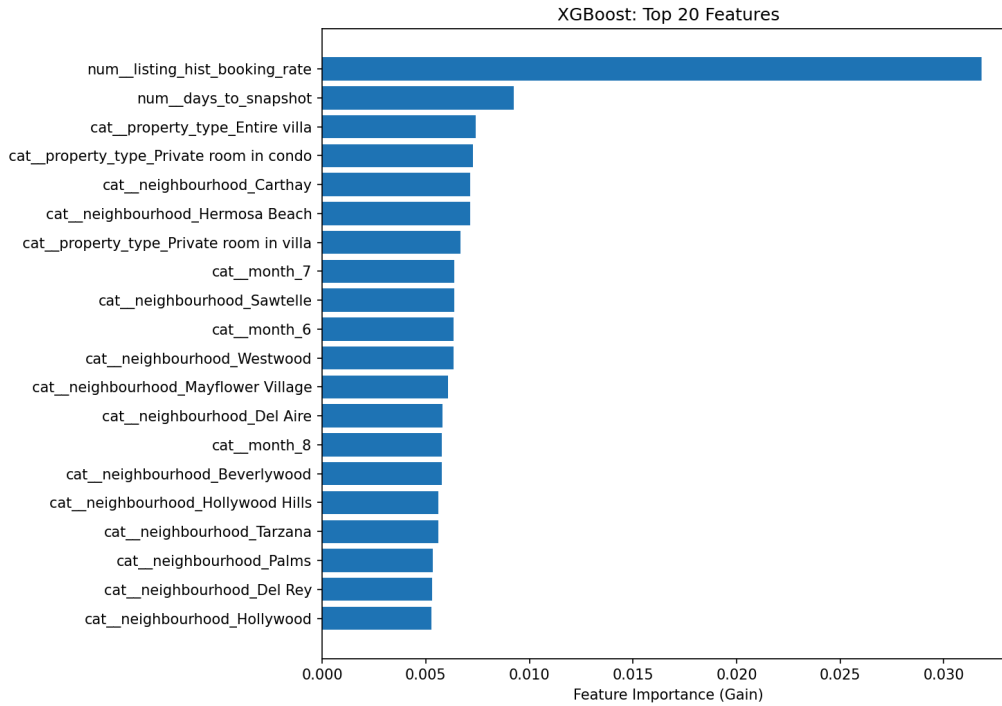
Appendix Figure A1. Confusion Matrices on the Test Set



Appendix Figure A2. Booking Rate by Days to Snapshot



Appendix Figure A3. XGBoost Top 20 Features



Appendix Table A1. Calibration by Prediction Decile

Decile	Predicted Prob.	Actual Rate	N	Model
0	0.0887	0.4413	24798	Logistic (L1)
1	0.1370	0.4388	24798	Logistic (L1)
2	0.1892	0.4342	24797	Logistic (L1)
3	0.2479	0.4267	24798	Logistic (L1)
4	0.3150	0.4109	24798	Logistic (L1)
5	0.3940	0.4136	24797	Logistic (L1)
6	0.4919	0.4402	24798	Logistic (L1)
7	0.6078	0.4842	24797	Logistic (L1)
8	0.7348	0.5288	24798	Logistic (L1)
9	0.8713	0.6014	24798	Logistic (L1)
0	0.0050	0.3729	24798	XGBoost
1	0.0285	0.3808	24798	XGBoost
2	0.0834	0.3880	24797	XGBoost
3	0.1836	0.3978	24798	XGBoost
4	0.3831	0.4787	24798	XGBoost
5	0.6994	0.4396	24797	XGBoost
6	0.8975	0.4295	24798	XGBoost
7	0.9667	0.5081	24797	XGBoost
8	0.9920	0.5568	24798	XGBoost
9	0.9988	0.6680	24798	XGBoost

Appendix Table A2. Event Robustness Diagnostics

Table A2.1 Event Signal Strength by Data Split

Block	Group	Metric	Value
signal_strength	Train	Unique Dates	305.0000
signal_strength	Validation	Unique Dates	172.0000
signal_strength	Test	Unique Dates	172.0000
signal_strength	Train	Event Dates	2.0000
signal_strength	Validation	Event Dates	108.0000
signal_strength	Test	Event Dates	142.0000
signal_strength	Train	Event-Day Share	0.0066
signal_strength	Validation	Event-Day Share	0.6279
signal_strength	Test	Event-Day Share	0.8256
signal_strength	Train	Mean Event Count (Total)	0.0328
signal_strength	Validation	Mean Event Count (Total)	10.0523
signal_strength	Test	Mean Event Count (Total)	3.3779
signal_strength	Train	Mean Event Count (City Calendar)	2.1443
signal_strength	Validation	Mean Event Count (City Calendar)	0.3140
signal_strength	Test	Mean Event Count (City Calendar)	0.0000
signal_strength	Train	Mean Event Count (TSE Permit)	8.6689
signal_strength	Validation	Mean Event Count (TSE Permit)	5.6221
signal_strength	Test	Mean Event Count (TSE Permit)	1.1279

Table A2.2 Baseline vs. Event Controls: Model Performance

Block	Model / Metric	Scenario	Value
baseline_vs_events_delta	L1 Logistic / ROC AUC	Baseline	0.5480
baseline_vs_events_delta	L1 Logistic / PR AUC	Baseline	0.5183
baseline_vs_events_delta	L1 Logistic / F1 Score	Baseline	0.5095
baseline_vs_events_delta	L1 Logistic / Brier Score	Baseline	0.2907
baseline_vs_events_delta	XGBoost / ROC AUC	Baseline	0.5936
baseline_vs_events_delta	XGBoost / PR AUC	Baseline	0.5666
baseline_vs_events_delta	XGBoost / F1 Score	Baseline	0.5696
baseline_vs_events_delta	XGBoost / Brier Score	Baseline	0.3625
baseline_vs_events_delta	L1 Logistic / ROC AUC	With Event Controls	0.5481
baseline_vs_events_delta	L1 Logistic / PR AUC	With Event Controls	0.5184
baseline_vs_events_delta	L1 Logistic / F1 Score	With Event Controls	0.5947
baseline_vs_events_delta	L1 Logistic / Brier Score	With Event Controls	0.2924
baseline_vs_events_delta	XGBoost / ROC AUC	With Event Controls	0.5902
baseline_vs_events_delta	XGBoost / PR AUC	With Event Controls	0.5663
baseline_vs_events_delta	XGBoost / F1 Score	With Event Controls	0.5592
baseline_vs_events_delta	XGBoost / Brier Score	With Event Controls	0.3564
baseline_vs_events_delta	L1 Logistic / ROC AUC	Change	0.0001
baseline_vs_events_delta	L1 Logistic / PR AUC	Change	0.0001
baseline_vs_events_delta	L1 Logistic / F1 Score	Change	0.0852
baseline_vs_events_delta	L1 Logistic / Brier Score	Change	0.0017
baseline_vs_events_delta	XGBoost / ROC AUC	Change	-0.0034
baseline_vs_events_delta	XGBoost / PR AUC	Change	-0.0004
baseline_vs_events_delta	XGBoost / F1 Score	Change	-0.0104
baseline_vs_events_delta	XGBoost / Brier Score	Change	-0.0061

Appendix Table A3. Weather Robustness Diagnostics

Table A3.1 Weather Signal Summary by Split Data

Block	Group	Metric	Value
weather_signal	Train	Unique Dates	305.0000
weather_signal	Validation	Unique Dates	172.0000
weather_signal	Test	Unique Dates	172.0000
weather_signal	Train	Non-Missing Share (Temp Max)	1.0000
weather_signal	Validation	Non-Missing Share (Temp Max)	1.0000
weather_signal	Test	Non-Missing Share (Temp Max)	1.0000
weather_signal	Train	Non-Missing Share (Temp Min)	1.0000
weather_signal	Validation	Non-Missing Share (Temp Min)	1.0000
weather_signal	Test	Non-Missing Share (Temp Min)	1.0000
weather_signal	Train	Non-Missing Share (Precipitation)	1.0000
weather_signal	Validation	Non-Missing Share (Precipitation)	1.0000
weather_signal	Test	Non-Missing Share (Precipitation)	1.0000
weather_signal	Train	Unique Values (Temp Max)	162.0000
weather_signal	Validation	Unique Values (Temp Max)	58.0000
weather_signal	Test	Unique Values (Temp Max)	7.0000
weather_signal	Train	Unique Values (Temp Min)	122.0000
weather_signal	Validation	Unique Values (Temp Min)	56.0000
weather_signal	Test	Unique Values (Temp Min)	7.0000
weather_signal	Train	Unique Values (Precipitation)	36.0000
weather_signal	Validation	Unique Values (Precipitation)	20.0000
weather_signal	Test	Unique Values (Precipitation)	7.0000

Table A3.2 Impact of Weather Controls on Model Performance

Block	Model / Metric	Scenario	Value
baseline_vs_weather_delta	L1 Logistic / ROC AUC	Baseline	0.5480
baseline_vs_weather_delta	L1 Logistic / PR AUC	Baseline	0.5183
baseline_vs_weather_delta	L1 Logistic / F1 Score	Baseline	0.5095
baseline_vs_weather_delta	L1 Logistic / Brier Score	Baseline	0.2907
baseline_vs_weather_delta	XGBoost / ROC AUC	Baseline	0.5936
baseline_vs_weather_delta	XGBoost / PR AUC	Baseline	0.5666
baseline_vs_weather_delta	XGBoost / F1 Score	Baseline	0.5696
baseline_vs_weather_delta	XGBoost / Brier Score	Baseline	0.3625
baseline_vs_weather_delta	L1 Logistic / ROC AUC	With Weather Controls	0.5477
baseline_vs_weather_delta	L1 Logistic / PR AUC	With Weather Controls	0.5181
baseline_vs_weather_delta	L1 Logistic / F1 Score	With Weather Controls	0.5091
baseline_vs_weather_delta	L1 Logistic / Brier Score	With Weather Controls	0.2909
baseline_vs_weather_delta	XGBoost / ROC AUC	With Weather Controls	0.5931
baseline_vs_weather_delta	XGBoost / PR AUC	With Weather Controls	0.5642
baseline_vs_weather_delta	XGBoost / F1 Score	With Weather Controls	0.5673
baseline_vs_weather_delta	XGBoost / Brier Score	With Weather Controls	0.3587
baseline_vs_weather_delta	L1 Logistic / ROC AUC	Change	-0.0002
baseline_vs_weather_delta	L1 Logistic / PR AUC	Change	-0.0002
baseline_vs_weather_delta	L1 Logistic / F1 Score	Change	-0.0004
baseline_vs_weather_delta	L1 Logistic / Brier Score	Change	0.0002
baseline_vs_weather_delta	XGBoost / ROC AUC	Change	-0.0005
baseline_vs_weather_delta	XGBoost / PR AUC	Change	-0.0025
baseline_vs_weather_delta	XGBoost / F1 Score	Change	-0.0023
baseline_vs_weather_delta	XGBoost / Brier Score	Change	-0.0038

Note: Supplementary tables, figures, and files are provided in the submitted ZIP archive.